

Printing, User Input, and Escape Sequences

Our First Interactive Programs



About Your Instructor

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Curriculum Design

Master of Science in Business Analytics (Hult)

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Emerging Jobs in 2024 (LinkedIn)

#1. Artificial Intelligence Specialist

Hiring growth: 74% annually
Skills: Machine Learning, Deep Learning, TensorFlow, Python, Natural Language Processing
LinkedIn Job postings: 1705 (in the USA)

#3. Data Scientist

Hiring growth: 37% annually
Skills: Machine Learning, Data Science, Python, R, Apache Spark
LinkedIn Job postings: 16468 (in the USA)

Role	AVG Salary (yearly)
Machine Learning Engineer	\$ 164,229
Data Scientist	\$ 135,056

Source: indeed.com

Learning to Code

Just like learning English, **learning a programming language takes time.**

We will progress step-by-step to build up your coding fluency.

Don't be Afraid to Paint



The Art of Coding

Coding is much more of an **art** than a **science**.

There are **many ways to create art**.

...and **it takes time** to perfect the vision you are trying to create.

Printing

purpose: outputting arguments to the console.

We **rarely** want to print everything. The **print()** command helps us manage what output we would like to display, and what output should be suppressed.

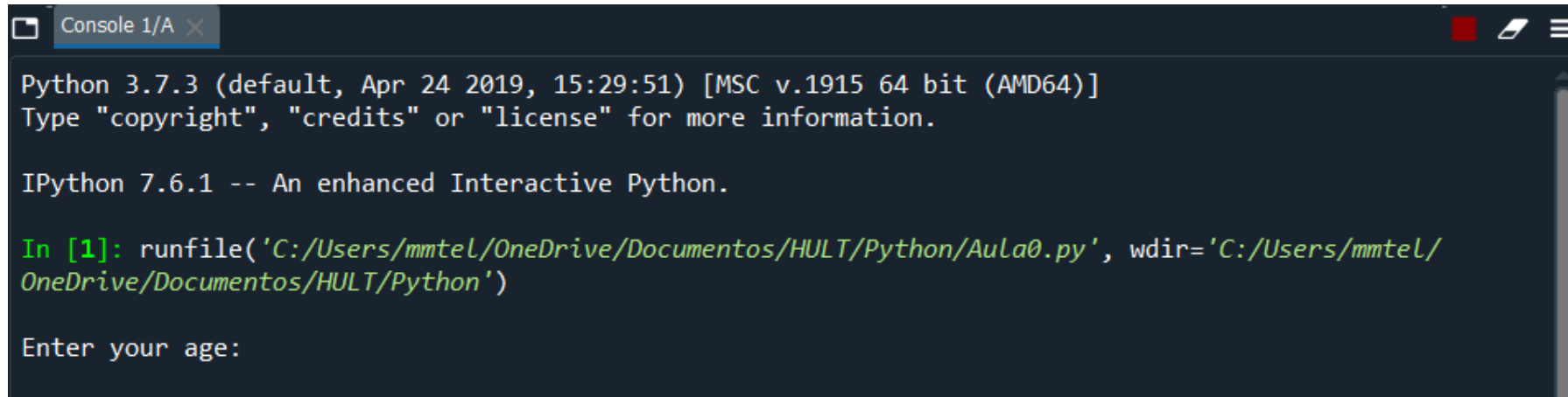
Printing

Essential `print( )` syntax

Command	Description
<code>print()</code>	outputs and argument to the console
<code>" "</code>	wrapper to print strings
<code>' '</code>	wrapper to print strings
<code>""" """</code>	wrapper to print strings on multiple lines
<code>f" {var} "</code>	dynamically calls variables/objects into print strings

User Input

Prompting Users



```
Python 3.7.3 (default, Apr 24 2019, 15:29:51) [MSC v.1915 64 bit (AMD64)]
Type "copyright", "credits" or "license()" for more information.

IPython 7.6.1 -- An enhanced Interactive Python.

In [1]: runfile('C:/Users/mmtel/OneDrive/Documentos/HULT/Python/Aula0.py', wdir='C:/Users/mmtel/
OneDrive/Documentos/HULT/Python')

Enter your age:
```

input()

purpose: allows us to prompt users to give us input.

We can save this input as a variable and use it later.

This allows for **interactive and dynamic programs**.

input()

```
x = input()
```

A user may have no idea they are being asked to do something.

input()

```
x = input(prompt = "> ")
```

>

```
print(x)
```

> *[prints a user's response]*

input()

```
x = input(prompt = "Response: ")
```

Response:

```
print(x)
```

Response: *[prints a user's response]*

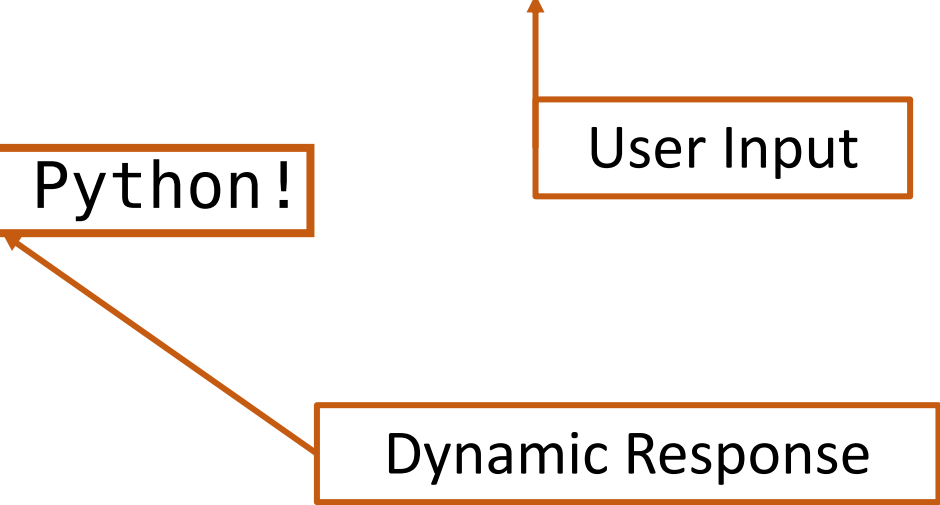
input()

```
user_name = input(prompt = "What's your name? ")  
print(f"Welcome {user_name}!")
```

> What's your name? Python

> Welcome Python!

User Input

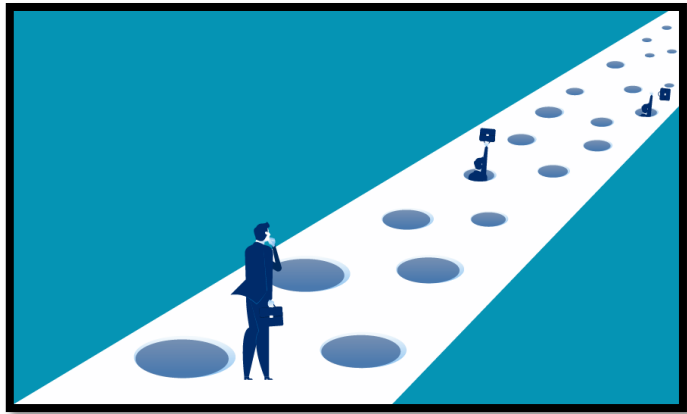


Dynamic Response

Type Errors with **input**



Potential Problems to Avoid



Type Errors with `input`

Variables saved from `input( )` are of type string (`str`)

```
val = input(prompt = "> ")
```

```
> 1
```

```
type(val)
```

str

Type Errors with `input`

We can alter this using wrappers

```
val = int( input(prompt = "> ") )
```

```
> 1
```

```
type(val)
```

int

Type Errors with `input`

We can alter this using wrappers

```
val = float( input(prompt = "> ") )
```

```
> 1
```

```
type(val)
```

float

Type Errors with `input`

We can alter this using wrappers

```
val = bool( input(prompt = "> ") )
```

```
> 1
```

```
type(val)
```

bool

Type Errors with `input`

Be Careful!

```
val = int( input(prompt = "> ") )
```

```
> eggplant
```

```
type(val)
```

ValueError